

Jyothi Swaroop Akasamu

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🔗 (Github link)

🌐 (Linkedin link)

Summary

AI/ML Engineer and Data Scientist specializing in the development and deployment of production-grade AI systems, agentic workflows, and scalable backend applications. Proven expertise in building production-level RAG frameworks, end-to-end automated ETL pipelines, and optimizing real-time data orchestration using Python, LangChain, and Flask. Strong background in engineering high-impact, localized AI solutions from concept to deployment within production environments.

Experience

- **AI-Data Scientist – Empover I-Tech** Onsite | OCT 2025 – Current
 - Architected an end-to-end ETL automation suite and data orchestration pipeline using Python and Agentic AI, reducing manual processing by 80% with 99.9% data integrity.
 - Implemented a multilingual RAG framework using LangChain, FAISS, and local LLMs, while deploying optimized Flask backend APIs for real-time systems.
- **Machine Learning Intern – Luein Analytics** Remote | JUN 2025 – SEP 2025
 - Deployed Django-based web applications integrated with predictive machine learning models, leveraging modular backend components for high scalability.
- **ML/Backend Intern – Renote.ai** Hybrid | FEB 2025 – MAY 2025
 - Developed a sketch-to-UI detection pipeline using Google AI Studio and Gemini LLM, translating hand-drawn wireframes into structured JSON layouts.
 - Engineered a text-to-video generation microservice utilizing Runway ML APIs, automating prompt handling.

Projects

- **Gesture Recognition for Human-Computer Interaction:** Built a real-time hand gesture recognition system using OpenCV and Scikit-learn, achieving 97% classification accuracy. Applied EDA, PCA, and hyperparameter tuning to optimize model performance for webcam-based interaction. [Link]
- **Mobile UI Element Detection from Sketches:** Built a pipeline using Google AI Studio to detect UI components from hand-drawn sketches and convert them into JSON layouts. Integrated Gemini LLM for prompt-based classification, enabling rapid UI prototyping. [Link]
- **AI-Powered Crop Protection Using EfficientNetB0:** Developed an image classification model using EfficientNetB0 to detect 12 types of crop pests with 86% accuracy. Deployed a Flask web interface enabling farmers to upload pest images and receive instant treatment suggestions. [Link]

Skills

- **Programming & Databases:** Python, SQL, NoSQL (MongoDB), Bash, Git
- **AI / Machine Learning:** Deep Learning, Generative AI, LLMs, RAG, LangChain, FAISS, TensorFlow, PyTorch, Scikit-learn, Hugging Face, NLP
- **Data Engineering & Analytics:** ETL Pipelines, Data Preprocessing, Feature Engineering, EDA, Pandas, NumPy, Statistics
- **Backend & DevOps:** Django, Flask, REST API Design, Docker, GitHub Actions, CI/CD, Postman, Swagger
- **Cloud Platforms:** GCP, AWS, Microsoft Azure
- **Soft Skills:** Problem Solving, Agile Development, Effective Communication, Languages Known- English, Hindi, Telugu

Education

- **Chandigarh University** Gharuan, Punjab
B.E. in Computer Science and Engineering (AI & ML) – GPA: 7.7/10.0 Sept 2021 – June 2025
- **Sri Chaitanya Junior College** Vijayawada, Andhra Pradesh
Intermediate (Class XII) – Score: 967/1000 Aug 2019 – May 2021

Achievements

- College ML Hackathon – Secured 2nd position for engineering a high-performance machine learning solution
- Cisco Sustainability Hackathon – Secured 5th place nationally for presenting an AI-driven sustainability prototype.